

Carry Trade: Evolution through years

International Finance: Literature Review & Strategy Analysis

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1 Introduction

The carry trade is considered a cornerstone strategy in the foreign exchange (FX) market: simple in concept yet enduringly effective. Investors borrow in currencies with low interest rates and invest in those with higher rates, thereby profiting from the interest rate differential. The theoretical foundation of the carry trade stems from the violation of the Uncovered Interest Rate Parity (UIP), which posits that differences in interest rates across countries should be offset by expected changes in exchange rates.

However, empirical studies (Fama, 1984), (Burnside et al., 2006) consistently demonstrate that the UIP condition fails in practice, creating persistent profitable opportunities through carry trades. The academic relevance of the carry trade lies not only in its challenge to established economic theories but also in its practical implications for portfolio construction, global asset allocation, and risk management. This literature review examines the evolution of carry trade research and practice, organizing the analysis into two periods: before the 2008 financial crisis, and during and after the crisis period.

2 Discovery of Carry Trade and its Success until 2008

Early research into the carry trade emerged from the observed empirical failures of the UIP condition. Fama (1984) was among the first to document this anomaly, noting that currencies with higher interest rates tend to appreciate rather than depreciate, thereby contradicting the expectations set by UIP. This paradox sparked a rich body of research aimed at uncovering possible explanations, including time-varying risk premia (Engel, 1996) and behavioral biases in investor expectations.

Throughout the early 2000s, the FX market experienced a surge in trading activity, driven by both institutional and retail investors. Empirical studies conducted prior to the 2008 financial crisis demonstrated that carry trades offered substantial excess returns, particularly among G10 currencies. Galati and Melvin (2004) highlighted the dramatic growth in FX trading volumes, attributing it to technological advancements and the increased globalization of capital markets. Burnside, Eichenbaum, Kleshchelski, and Rebelo (2006) further demonstrated that simple carry trade strategies could achieve Sharpe ratios comparable to those of equities, but with relatively low volatility.

The resilience and profitability of carry trades during this period puzzled many economists, raising debates over whether these returns represented true arbitrage opportunities or compensation for hidden risks. As FX trading infrastructure evolved, implementing strategies such as the carry trade became increasingly accessible. King, Osler, and Rime (2013) detailed the rising influence of hedge funds and algorithmic trading, further institutionalizing the carry trade as a mainstream investment strategy.

Despite their historical profitability, carry trades are associated with “crash risk” or sudden, large losses associated with reversals in currency values. Brunnermeier, Nagel, and Pedersen (2008) highlighted that carry trades are susceptible to funding constraints and investor herding, which can exacerbate unwinding events during times of market stress. They emphasized that during low-

volatility periods, traders become over-leveraged, making the strategy prone to abrupt crashes when volatility returns. Moreover, the literature suggested that carry returns were correlated with global risk appetite, particularly during times of financial tightening or geopolitical instability (Lustig & Verdelhan, 2007).

3 Performance of the Carry Trade During and After the 2008 Financial Crisis

The carry trade relies on the assumption that profits from interest rate differentials will not be neutralized by adverse exchange rate movements, thereby implying a persistent violation of uncovered interest parity. However, historical evidence indicates that this assumption often breaks down under conditions of financial stress.

Prior to the financial crisis of 2007–2008, ‘safe haven currencies’ borrowed for the purposes of this trade were not only low-yielding, but their issuing countries were characterised by political and financial stability, and a net foreign asset position (McCauley & McGuire, 2009). This instilled trust in international investors. Prominent examples at the time included Japan and Switzerland, or the Japanese yen (JPY) and Swiss franc (CHF), both seen as low-risk.

Carry trades tend to fail due to two factors: high risk aversion and high volatility in the foreign exchange market (Falconio, 2016). In the wake of the US subprime loan crisis, investors began to question the quality of certain asset classes, particularly fixed income and equities. Triggering widespread turmoil in global financial markets, this loss in investor confidence led to contagion in the FX market, thereby affecting the carry trade. Furthermore, risk aversion dramatically increased, due in part to an increase in illiquidity, as some fund managers froze withdrawals when investors tried to pull out of risky investments. Large, previously profitable interest rate differentials were no longer sufficient to compensate the risk of currency loss.

In August 2007, market volatility led to the inversion of the yields of currencies used in carry, resulting in the first wave of poor carry returns. For instance, the JPY/AUD exchange rate saw a variation of 7.7% on August 15th, 2007, in contrast to a daily average of 0.7% up until that point (Melvin & Taylor, 2009). When volatility peaked in late 2008, with a VIX index of 80 on October 27th, the currencies whose yield had been the highest in prior months saw the most depreciation against the USD (McCauley & McGuire, 2009).

Empirical data illustrate the dramatic decline in carry trade performance: while the average excess return on uncovered carry trades stood at 3.41% prior to the crisis, it plummeted to 0.73% in its aftermath (Fan, Paseka, Qi, & Zhang, 2022). Capital flows during the crisis period further strained the viability of the carry trade. Significant capital flight from East Asian economies precipitated currency depreciations relative to the USD, eroding the profitability of investments in high-yielding currencies. Concurrently, in response to the economic downturn, central banks around the world—most notably the U.S. Federal Reserve—implemented aggressive monetary easing, including bringing policy rates close to the zero lower bound and initiating quantitative easing programs.

Although these actions widened interest rate differentials once again, persistent risk aversion among investors impeded a full recovery of carry trade profitability. The post-crisis environment revealed that while funding conditions were favorable, risk dynamics had fundamentally shifted, requiring a reevaluation of traditional carry strategies.

4 Conclusion

The carry trade has long challenged the foundations of international finance, revealing persistent deviations from uncovered interest parity and offering investors attractive, albeit risky, return opportunities. Before 2008, the strategy was notably profitable, benefiting from global liquidity, low volatility, and relatively stable market conditions. However, the global financial crisis exposed its vulnerabilities, demonstrating how increased risk aversion and market stress could swiftly reverse carry trade gains. Although monetary policy responses initially restored favorable funding conditions, lingering uncertainty and evolving risk factors have reshaped the landscape for FX investors.

Today, the FX market faces new complexities, including divergent monetary policies, increased geopolitical tensions, and structural changes. With the arrival of alternative means of payment, the crypto carry trade has emerged: traders borrow in low-yielding fiat currency and invest in high-yield cryptocurrency savings accounts. This shift reflects broader market innovation. The study by Koijen, Moskowitz, Pedersen, and Vrugt (2018) has already broadened the scope of the carry trade concept to other asset classes like global equity, commodity, bond, options, and other markets.

Thus, looking ahead, the future of carry trades presents even greater potential and uncertainty. With continued technological advancements, evolving market structures, and the increasing integration of digital assets like cryptocurrencies, the carry trade is likely to evolve in ways that we are only beginning to understand. As interest rate differentials persist, but new liquidity and risk factors emerge, the opportunities—and risks—associated with carry trades will likely expand, demanding new strategies and more sophisticated risk management tools from investors.

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1 Carry Trade Strategy Analysis

1.1 Excess return

The monthly average return is 0.66%, translating to an annualized return of 7.95%. The annualized standard deviation (0.0804) indicates relatively high volatility, though the data exhibits mean reversion. The skewness (0.0864) suggests a slight right skew, meaning extreme positive returns occur more frequently than extreme negative ones. The kurtosis (2.94) is close to 3, implying a near-normal distribution.

Total Return	Daily Return	Trading Days
160,28%	0,0318%	250

Table 1: Performance Summary

Mean	Standard Deviation	Skewness	Kurtosis
0,66%	0,0232	0,2994	2,3147

Table 2: Montly statistics

Mean	Standard Deviation	Skewness	Kurtosis
7,95%	0,0804	0,0864	2,9429

Table 3: Annual statistics



Figure 1: Excess returns

1.2 Spot return

In contrast, the spot return has a negative annualized mean (-7.58%), indicating that carry trade strategies underperform in spot returns compared to excess returns. The standard deviation (0.0772) is slightly lower than for excess returns, suggesting fewer extreme movements. The

skewness (0.058) remains positive but less pronounced, meaning extreme positive returns still marginally outweigh negative ones on average. Like excess returns, kurtosis (2.93) is near normal.

Total Return	Daily Return	Trading Days
-154,83%	-0,0303%	250

Table 4: Performance Summary

Mean	Standard Deviation	Skewness	Kurtosis
-0,63%	0,0223	0,2033	2,2116

Table 5: Monthly statistics

Mean	Standard Deviation	Skewness	Kurtosis
-7,58%	0,0772	0,0587	2,9343

Table 6: Annual statistics

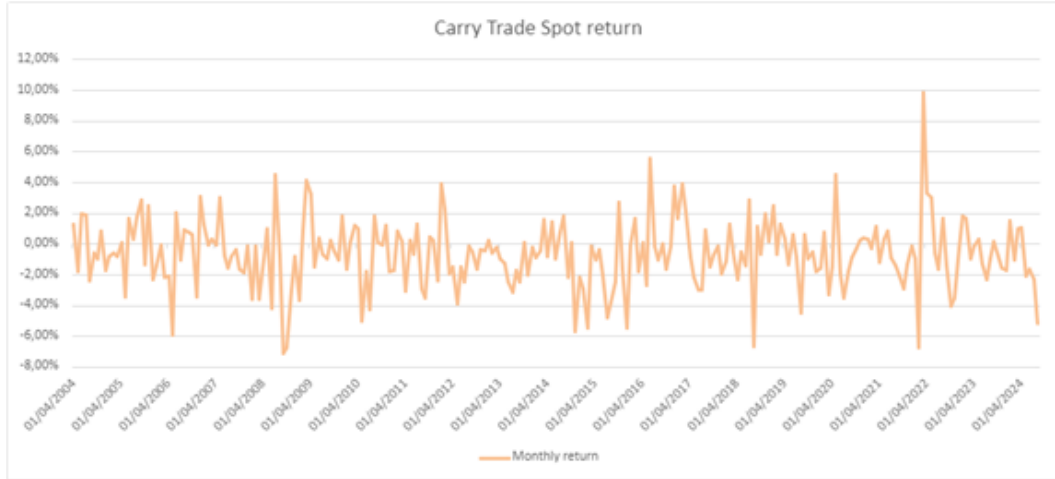


Figure 2: Spot returns

2 Momentum Strategy Analysis

2.1 Spot return

The strategy delivers strong performance in spot returns, with a monthly mean of 0.88% (annualized 10.59%). The standard deviation (0.0543) indicates relatively low volatility compared to carry trade strategies. The skewness (0.2476) is more pronounced than in carry trade returns, suggesting a greater tendency for extreme positive returns. The kurtosis (3.19) exceeds 3, implying a slightly fatter-tailed distribution than normal, with more frequent extreme values.

Total Return	Daily Return	Trading Days (approximate)
216,14%	0,0423%	250

Table 7: Performance Summary

Mean	Standard Deviation	Skewness	Kurtosis
0,88%	0,0157	0,8577	5,3276

Table 8: Monthly statistics

Mean	Standard Deviation	Skewness	Kurtosis
10,59%	0,0543	0,2476	3,1940

Table 9: Annual statistics

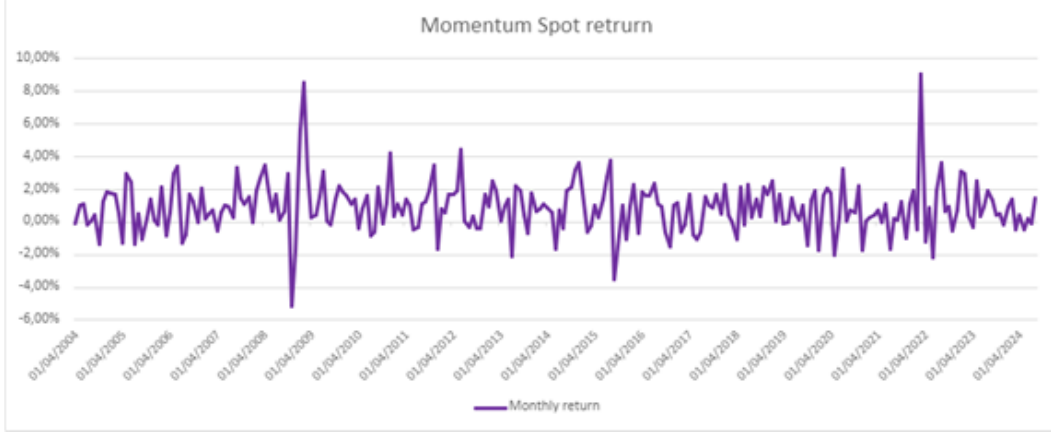


Figure 3: Spot returns

2.2 Excess return

The excess returns underperform, with a negative annualized mean (-1.74%) and higher volatility (standard deviation: 0.0624). The skewness (0.1169) remains positive, indicating a longer right tail, though less extreme than in spot returns. The kurtosis (3.11) also suggests a mild deviation from normality, with marginally more extreme observations.

Total Return	Daily Return	Trading Days (approximate)
-35,46%	-0,0069%	250

Table 10: Performance Summary

Mean	Standard Deviation	Skewness	Kurtosis
-0,14%	0,0180	0,4050	4,3255

Table 11: Monthly statistics

Mean	Standard Deviation	Skewness	Kurtosis
-1,74%	0,0624	0,1169	3,1105

Table 12: Annual statistics

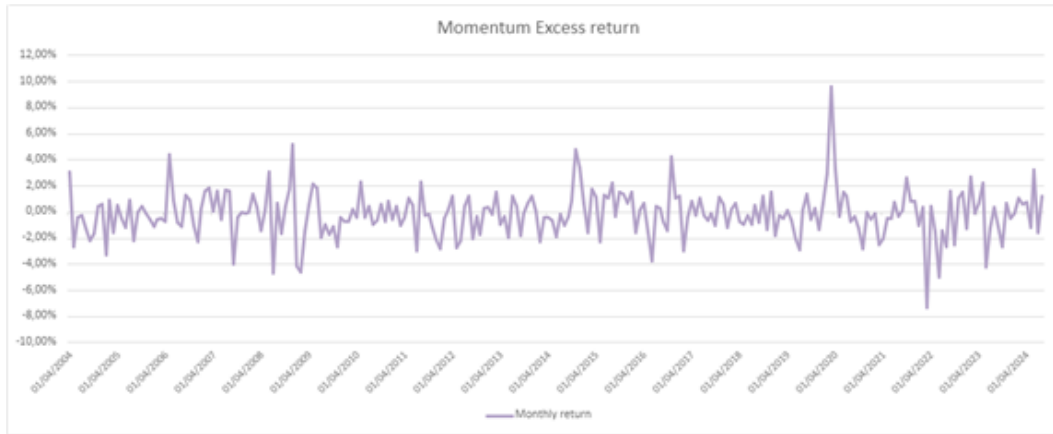


Figure 4: Excess returns

3 Conclusion

Performance Comparison:

- Carry Trade: Effective in excess returns (annualized 7.95%) but ineffective in spot returns (-7.58%), aligning with the Fama test on market efficiency.
- Momentum Strategy: Strong in spot returns (10.59%) but weak in excess returns (-1.74%).

Graphical Insights:

Carry Trade: Excess and spot returns move in the same direction, but excess returns are systematically higher. Spot returns exhibit fewer extreme negatives, yet their cumulative effect drives the overall mean downward.



Figure 5: Carry Trade

Momentum Strategy: Spot and excess returns frequently move in opposite directions, reflecting a negative correlation—a key divergence from carry trade dynamics.

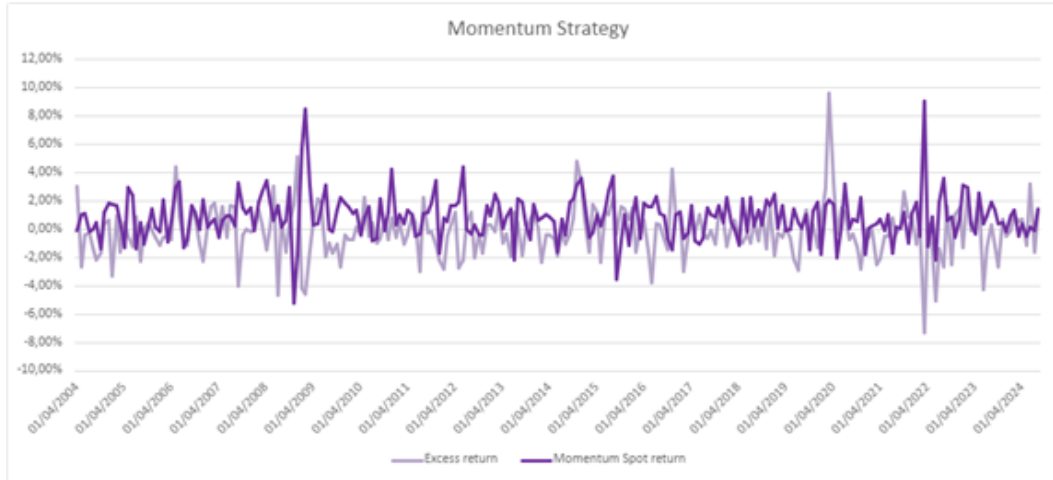


Figure 6: Momentum Strategy

These findings support the theoretical expectation that carry trade profits arise from interest rate differentials (captured in excess returns), while momentum gains stem from spot price trends.

Subprime Crisis:

The lowest monthly excess return of the carry trade (-6.16%) was on the 1st septembre 2008 and the lowest monthly spot return of the momentum strategy (-5.16%) was on the 1st of november 2008, both happened during the subprime crisis.

The impact of the subprime crisis is remarkable for both carry and momentum. The fall of carry returns starts the 1/07/2008 and on the 1/09/2008 we observe the lowest value of -6.16%. This can be explained by a massive unwinding of carry trades: risk appetite fell down, the currency portfolios was deleveraged because of massive losses in bond & equity portfolios, investors were rushed to unwind (to buy back) theirs carry trades because of the uncertainty on the market and this contributed to strengthening up of usually low-yielding currencies (JPY, CHF and others). And as we know, a carry trade can be profitable as long as the investment currency does not adjust against the funding currency, so in the context of appreciation of the “safe” currencies, a carry trade is no more effective. Other factors like extreme volatility, funding liquidity constraints and so one have amplified the situation.

The momentum strategy spot returns collapsed during October 2008 until achieving its lower value of -5.16%. The essential problem was that momentum signals broke down because trends were violently and fast reversing. In the situation of very high volatility, when all assets started moving together making diversification almost impossible, only a high frequency momentum strategy could be profitable. In our case, we are doing monthly reequilibration and that is too slow to adjust to the market changes during the crisis.

Interesting to notice that 3 months after the lowest value of the momentum strategy, we have a pick at 8.51% on the 1/02/2009 and a fall down next month at 3.17% and the month after to 0.30%. So in theory we can adjust the strategy to the new trends, but then our returns will collapse again when the market starts to go back to its normal state. Carry trade has adjusted at

1/03/2009 with a pick at 7.54%, but then has fallen the two next month to -0.14%.

Extreme value in 2018:

On 1/08/2018 the carry trade excess return fell to its 4th lowest extreme value -5.92%. We think it can be linked to the president Donald Trump's trade war with China and the slowdown in global economic growth. There was no significant impact on the momentum spot return in the same period perhaps because the spot rate had adjusted better to the economic context and the signals were staying correct.

COVID 2019 - Period 2019-2020:

Unlike the 2008 crisis, the COVID-19 pandemic did not produce extreme negative returns in either carry trade or momentum strategies. The carry trade generated an average annual excess return of 3.78%, while momentum delivered 7.1% in spot returns—both lower than their long-term averages. This underperformance can be attributed to heightened volatility, which disrupted the predictability of spot and forward rate trends. However, the dynamics differed sharply from the 2008 crisis: COVID-19 was a global shock that simultaneously affected both high- and low-yield currencies, diluting the asymmetric currency movements that typically drive carry trade profits. Momentum strategies, meanwhile, faced erratic trends due to rapid policy shifts (e.g., central bank interventions) and synchronized asset price swings, reducing the efficacy of trend-following signals.

War in the Ukraine 2022:

The outbreak of the Russia-Ukraine war in February 2022 initially triggered sharp losses, with carry trade excess returns plunging to -5.98% and momentum spot returns dropping to -0.48% by 1 February 2022, reflecting panic-driven volatility and Rouble depreciation. However, by March 2022, both strategies rebounded dramatically—carry trade excess returns peaked at 11.65%, and momentum spot returns surged to 9.08%. This seems strange to us at first glance. However, as we assume that there are no liquidity limitations to trade, it is more reasonable that we have peaks of return in two strategies. As Western sanctions froze Russian assets, the Rouble's artificial depreciation (and subsequent central bank-driven recovery) created exploitable mispricings in forward rates (benefiting carry trades) and violent trend reversals (aiding momentum). In reality, these theoretical gains were inaccessible due to capital controls and trading restrictions on the Rouble, underscoring the limits of backtested models during geopolitical crises.